Triple M

Nov,2025

CCNA PROJECT

Implementation and customization our company network

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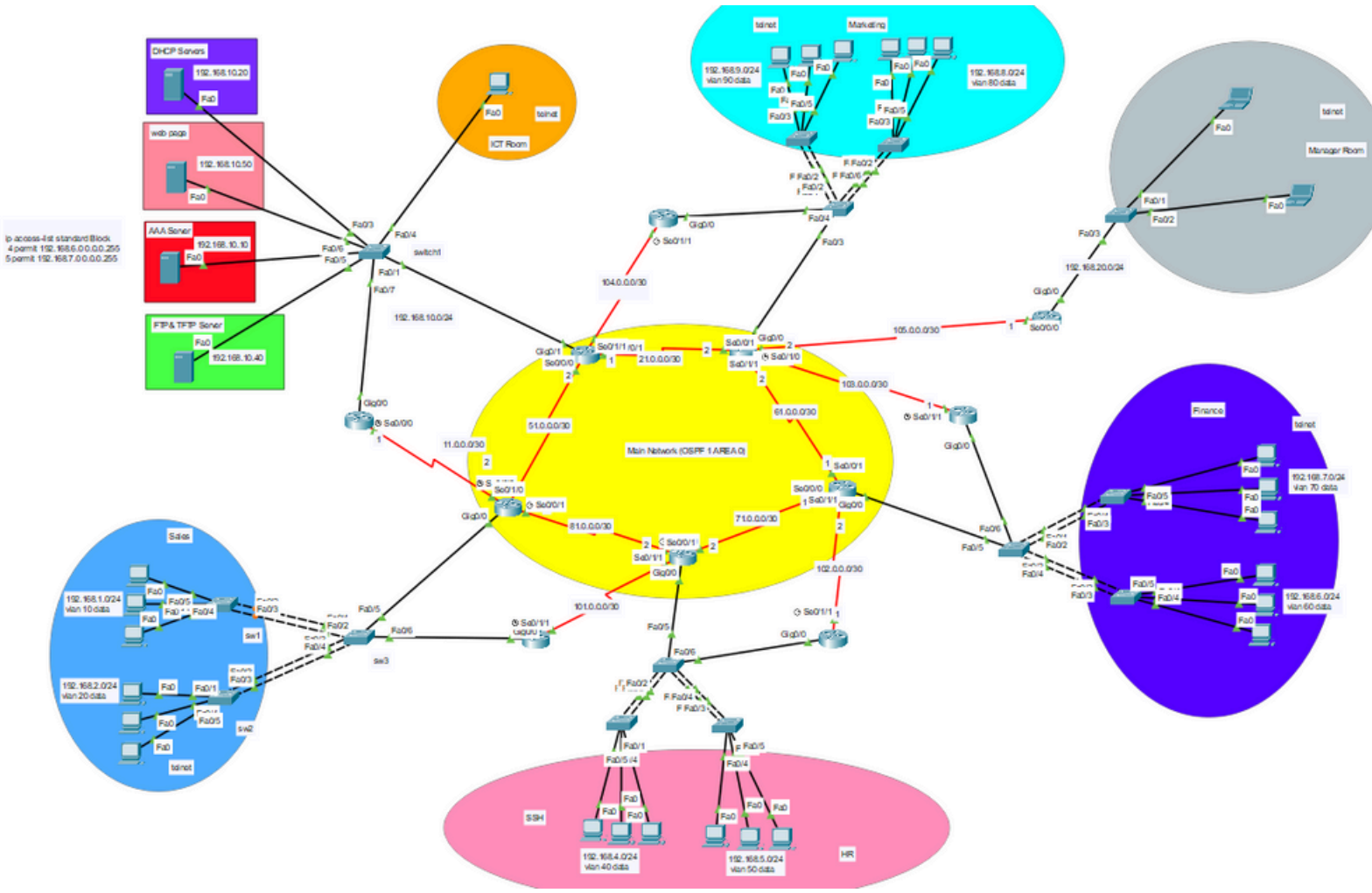


Introduction

“

In this network topology,
I have several departments such as **Sales**, **Marketing**, **HR**, **Finance**, **ICT**, and **Manager**.

Each department can only communicate within itself, while the Manager and ICT departments can communicate with any other department in the company.

We will also explain how this setup was implemented, highlight the advantages of the design, describe the **servers** and **multiple backup systems** in place, and emphasize **the high level of security** integrated into the network.

Routing Protocol

In the project, we used two excellent protocols, **OSPF** and **Static**,
which greatly helped us achieve the goal of this topology ,

- **OSPF** (**O**pen **S**hortest **P**ath **F**irst):
This is a dynamic protocol that uses the shortest path algorithm to determine the best route between routers in the network. It is fast in updates and handles large and complex networks efficiently.

  - **Configuration:**

```
router ospf 1
 log-adjacency-changes
 network 192.168.1.0 0.0.0.255 area 0
 network 192.168.2.0 0.0.0.255 area 0
 network 81.0.0.0 0.0.0.3 area 0
 network 51.0.0.0 0.0.0.3 area 0
 network 11.0.0.0 0.0.0.3 area 0
!
```

- **Static Routing** : This is a type of routing where paths are manually configured. It is simple and secure, but requires manual adjustment if there are changes in the network.

  - **Configuration:**

```
ip route 0.0.0.0 0.0.0.0 105.0.0.2
```

default route for manager's network

Server Setup

DHCP Server

- The goal of the **DHCP** Server in the project was to provide each department with a pool of IPs.
- This greatly helped in simplifying IP distribution across the network.
- The DHCP server assigned addresses to all devices except critical ones, such as:
  - **Servers** .
  - **ICT** room devices.
  - **Manager's** devices.
  - **Configuration:**

IP pools

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server
serverPool-50	192.168.5.1	0.0.0.0	192.168.5.5	255.255.255.0	10	0.0.0.0
serverPool-40	192.168.4.1	0.0.0.0	192.168.4.4	255.255.255.0	10	0.0.0.0
serverPool-70	192.168.7.1	0.0.0.0	192.168.7.7	255.255.255.0	10	0.0.0.0
serverPool-60	192.168.6.1	0.0.0.0	192.168.6.6	255.255.255.0	10	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	192.168.10.0	255.255.255.0	512	0.0.0.0

  - IP helper on routers to guide end users

```
ip helper-address 192.168.10.10
```

Server Setup

AAA Server

(**A**uthentication, **A**uthorization, and **A**ccounting) is used to :

- Control network access and manage user permissions.
- It verifies the user's identity before granting access and defines what actions or resources the user is allowed to use.
- It also tracks and logs user activities for monitoring and security purposes.

In the project, the **AAA** Server enhanced network security and ensured that only authorized users could access the **routers**

- **Configuration example on router Sales :**

```
aaa new-model
!
aaa authentication login jimmy group tacacs+ local
!
tacacs-server host 192.168.10.10 key 123
!
line vty 0 4
 login authentication jimmy
 transport input telnet
!
```

Server Setup

AAA Server

- **Telnet login example on routers Sales & Finance:**

```
Cisco Packet Tracer PC Command Line 1.0
C:\>telnet 192.168.2.2
Trying 192.168.2.2 ...Open


User Access Verification

Username: admin
Password:
Sales>
Sales>
```

```
C:\>telnet 192.168.7.3
Trying 192.168.7.3 ...Open


User Access Verification

Username: asd
Password:
Finance>
Finance>
Finance>
```

Username	Password
asd	asd
admin	123

Server Setup

Web Server

- We created a **web** server that all departments and users can access using the **public IP** “50.50.50.2”.
- The server also has a specific **private IP** assigned to it inside the network.
- we use **static NAT** to map this **private IP** to the **public ip**
    - **Configuration :**

```
ip nat inside source static 192.168.10.50 50.50.50.2
```

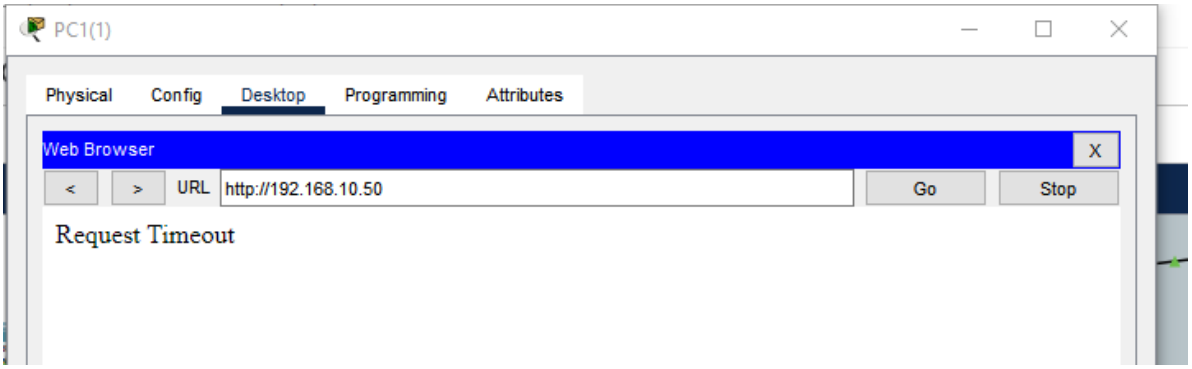
```
interface GigabitEthernet0/1
 ip address 192.168.10.200 255.255.255.0
 ip nat inside
 duplex auto
```

```
interface Serial0/0/0
 ip address 51.0.0.2 255.255.255.252
 ip nat outside
!
interface Serial0/0/1
 ip address 21.0.0.2 255.255.255.252
 ip nat outside
 clock rate 2000000
!
```

Server Setup

Web Server

- **Web Server Test :**



Server Setup

FTP & TFTP Server

- **FTP** (File Transfer Protocol):
  - Used to transfer files between devices over a network.
  - Supports authentication with a username and password.
  - Suitable when security and control over file transfers are needed.
- **TFTP** (Trivial File Transfer Protocol):
  - A simpler version of **FTP**, used to transfer small files like configurations or IOS images.
  - No authentication, making it faster but less secure.
  - Commonly used inside local networks or for configuring network devices.

- In our project, we used both **FTP** and **TFTP** to handle file transfers efficiently.

Remote Access (SSH & Telnet)

SSH (Secure Shell)

- A modern and secure protocol that uses encryption for all communications.
- Used to remotely manage devices in a safe and protected way.
- It is the preferred alternative to Telnet in security-focused networks.

- **Configuration example :**

we defined an enable secret password and two users with encrypted secrets to activate SSH on router HR

```
enable secret 5 $1$mERr$cgiXfUUSwjJUbgSZ9MEjw1

username asd secret 5 $1$mERr$1A6WmlgXdpa7HsGQy66Qd1
username mmm secret 5 $1$mERr$cgiXfUUSwjJUbgSZ9MEjw1
```

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ssh -l asd 192.168.5.3

Password:
HR>
HR>
```

Try username “asd” and password “asd”to test

Remote Access (SSH & Telnet)

Telnet

- An older protocol used to remotely access and manage network devices.
- All data is sent without encryption, which makes it less secure.
- Easy to use but not suitable for networks that require high security.

In our project,
We used **Telnet** with the **AAA Server**, which **provided authentication for secure access**.
We also used **SSH** for communication between the **ICT PCs and the routers**, ensuring a more secure and encrypted connection.

VLANS &VTP& R-O-A-S

VLANS

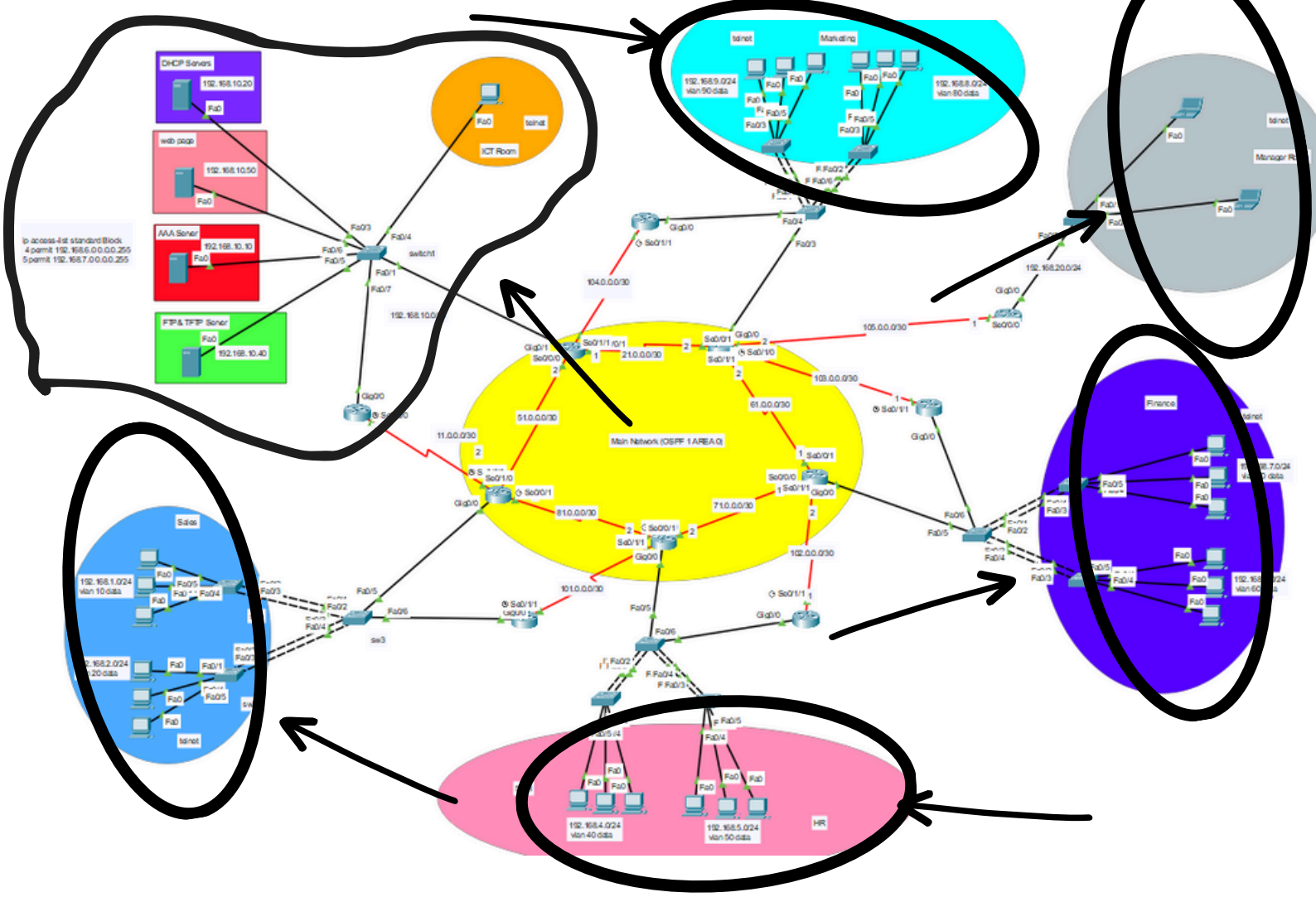
- We implemented **VLANs** (**V**irtual **L**ocal **A**rea **N**etworks) to logically separate the network into different segments.

- Each VLAN has its own subnet, which helps in **organizing** and managing the network efficiently.

- Having **multiple VLANs** provides several **Advantages:**
  - Improved network security by **isolating** departments from each other.

  - Better traffic management and **reduced** broadcast domains.

  - Easier network administration and **troubleshooting**.
- <u>Our project includes :</u>
  - A large number of **VLANs**, each with a distinct subnet to meet the needs of different departments.



Arrows are represent the locations of used **VLANS** in our project

VLANS &VTP& R-O-A-S

R-O-A-S

- **ROAS** (**R**outer **o**n **a S**tick) :
  is a network setup where a single router interface is used to route traffic between multiple VLANs.

- The router interface is configured as a trunk port to carry traffic for all VLANs.
- Each VLAN is assigned a **sub-interface** on the router with its own IP address for routing.

- This setup **allows inter-VLAN** communication without needing multiple physical router interfaces.

- **ROAS** is cost-effective and simplifies the network design while supporting multiple VLANs.

VLANS &VTP& R-O-A-S

R-O-A-S

- **Configuration example :**

Encapsulate all **VLANs** on the router interface by using a sub-interface for each **VLAN** configured in the network .

NOTE :
Don't forget to configure switch ports that carry more than one VLAN's traffic as **trunk** ports .

```
interface GigabitEthernet0/0
 no ip address
 ip ospf 1 area 0
 duplex auto
 speed auto
!
interface GigabitEthernet0/0.10
 no ip address
!
interface GigabitEthernet0/0.40
 encapsulation dot1Q 40
 ip address 192.168.4.3 255.255.255.0
 ip helper-address 192.168.10.20
 ip access-group Block out
 standby 3 ip 192.168.4.1
 standby 3 priority 150
 standby 3 preempt
!
interface GigabitEthernet0/0.50
 encapsulation dot1Q 50
 ip address 192.168.5.3 255.255.255.0
 ip helper-address 192.168.10.20
 ip access-group Block out
 standby 3 ip 192.168.5.1
 standby 3 priority 150
 standby 3 preempt
!
```

VLANS &VTP& R-O-A-S

VTP

- **VTP** (**V**LAN **T**runking **P**rotocol) is :
  a Cisco protocol used to manage VLANs across multiple switches.

- It allows **VLAN** information to be shared and synchronized automatically between switches in **the same VTP domain**.

- Helps in **reducing manual configuration**, since you can create or delete VLANs on one switch (the **VTP** server) and they are automatically updated on others.

- **VTP** works with trunk links, which carry multiple VLANs between switches.

- **Our project** used **VLANs** with **VTP** to make configuration easier and more organized.

VLANS &VTP& R-O-A-S

VTP

Server Switch Example :

```
Sales_mid#sh vtp status
VTP Version capable             : 1 to 2
VTP version running             : 1
VTP Domain Name                 : iti.com
VTP Pruning Mode                : Disabled
VTP Traps Generation            : Disabled
Device ID                       : 0001.6362.6520
Configuration last modified by 0.0.0.0 at 3-1-93 01:18:50
Local updater ID is 0.0.0.0 (no valid interface found)

Feature VLAN :
--------------
VTP Operating Mode                : Server
Maximum VLANs supported locally   : 255
Number of existing VLANs          : 8
Configuration Revision            : 5
MD5 digest                        : 0xC2 0xBE 0x82 0x10 0xD2 0x4B 0x4A 0xF8
                                    0x15 0xAF 0x7B 0x4B 0x67 0x27 0x4C 0xD5
Sales_mid#sh vtp password
VTP Password: mmm
Sales_mid#
```

```
VLAN Name                             Status
---- -------------------------------- --------
1    default                          active




10   data_sales                       active
20   VLAN0020                         active
100  voice                            active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

VLANS &VTP& R-O-A-S

VTP

- Client Switch Example :

```
Sales_up(config)#do sh vtp status
VTP Version capable             : 1 to 2
VTP version running             : 1
VTP Domain Name                 : iti.com
VTP Pruning Mode                : Disabled
VTP Traps Generation            : Disabled
Device ID                       : 0001.978B.AC20
Configuration last modified by 0.0.0.0 at 3-1-93 01:18:50

Feature VLAN :
--------------
VTP Operating Mode                : Client
Maximum VLANs supported locally   : 255
Number of existing VLANs          : 8
Configuration Revision            : 5
MD5 digest                        : 0xC2 0xBE 0x82 0x10 0xD2 0x4B 0x4A 0xF8
                                    0x15 0xAF 0x7B 0x4B 0x67 0x27 0x4C 0xD5
Sales_up(config)#do sh vtp password
VTP Password: mmm
Sales_up(config)#
```

```
Sales_up(config)#do sh vlan

VLAN Name                             Status
---- -------------------------------- --------
1    default                          active




10   data_sales                       active
20   VLAN0020                         active
100  voice                            active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

Backups (Eth.Channel & HSRP)

Ether-Channel

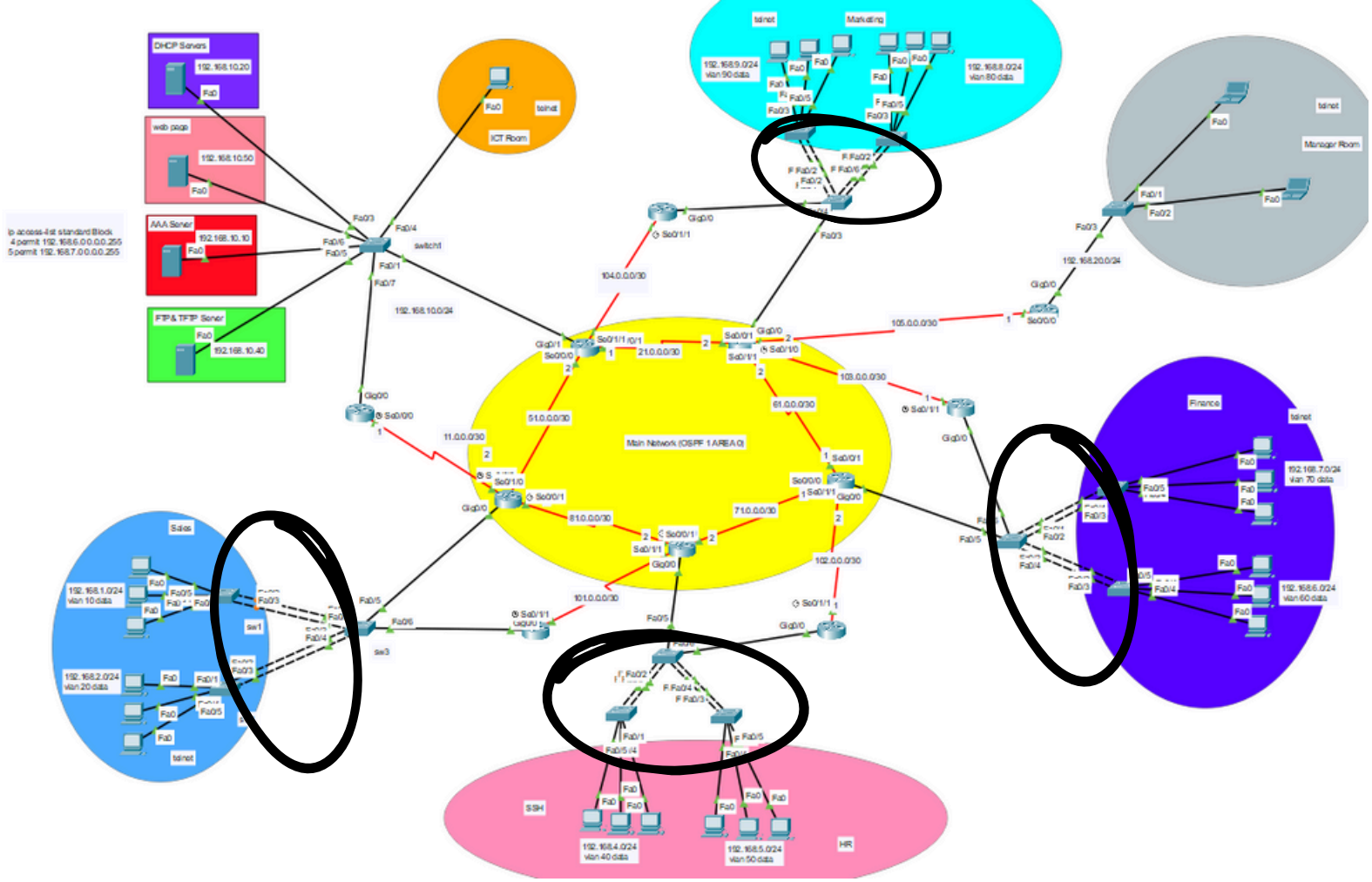
- **EtherChannel** is a technology that allows combining multiple physical links between switches or devices into a single logical link.
- We used it extensively in our project, which helped with load balancing across the network.
- It **increases bandwidth** by aggregating the capacity of all member links.
- Provides **redundancy**: if one link fails, the others continue to carry traffic.
- We used the Cisco-specific type (**PAgP**) for automatic configuration and negotiation.
- Helps in **reducing network congestion** and improving overall performance.

Backups (Eth.Channel & HSRP)

Ether-Channel

- **Configuration example :**

```
!
interface Port-channel5
 switchport trunk allowed vlan 60,70,100
 switchport mode trunk
!
interface Port-channel6
 switchport trunk allowed vlan 60,70,100
 switchport mode trunk
!
interface FastEthernet0/1
 switchport trunk allowed vlan 60,70,100
 switchport mode trunk
 channel-group 6 mode active
!
interface FastEthernet0/2
 switchport trunk allowed vlan 60,70,100
 switchport mode trunk
 channel-group 6 mode active
!
interface FastEthernet0/3
 switchport trunk allowed vlan 60,70,100
 switchport mode trunk
 channel-group 5 mode active
!
interface FastEthernet0/4
 switchport trunk allowed vlan 60,70,100
 switchport mode trunk
 channel-group 5 mode active
!
```



Ellipses are represent the locations of used **Ether-Channels** in our project

Subnetting Calculations

	Networks	Network ID / Mask	First Usable IP	Last Usable IP	Broadcast IP
1	sales vlan 10 : 20 Hosts	192.168.1.0 /27	192.168.1.1	192.168.1.30	192.168.1.31
2	sales vlan 20 : 20 Hosts	192.168.1.32 /27	192.168.1.33	192.168.1.62	192.168.1.63
3	HR vlan 40 : 20 Hosts	192.168.1.64 /27	192.168.1.65	192.168.1.94	192.168.1.95
4	HR vlan 50 : 20 Hosts	192.168.1.96 /27	192.168.1.97	192.168.1.126	192.168.1.127
5	Finance vlan 60 : 20 Hosts	192.168.1.128 /27	192.168.1.129	192.168.1.158	192.168.1.159
6	Finance vlan 70 : 20 Hosts	192.168.1.160 /27	192.168.1.161	192.168.1.190	192.168.1.191
7	Marketing vlan 80 : 20 Hosts	192.168.1.192 /27	192.168.1.193	192.168.1.222	192.168.1.223
8	Marketing vlan 90 : 20 Hosts	192.168.1.224 /27	192.168.1.225	192.168.1.254	192.168.1.255
9	Managers are 5 Hosts	192.168.2.0 /29	192.168.2.1	192.168.2.6	192.168.2.7
10	Router Link 1	192.168.2.8 /30	192.168.2.9	192.168.2.10	192.168.2.11

Subnetting Calculations

11	**Router Link 2**	192.168.2.12 /30	192.168.2.13	192.168.2.14	192.168.2.15
12	Router Link 3	192.168.2.16 /30	192.168.2.17	192.168.2.18	192.168.2.19
13	Router Link 4	192.168.2.20 /30	192.168.2.24 /30	192.168.2.22	192.168.2.23
14	Router Link 5	192.168.2.24 /30	192.168.2.25	192.168.2.26	192.168.2.27
15	Router Link 6	192.168.2.28 /30	192.168.2.29	192.168.2.30	192.168.2.31
16	Router Link 7	192.168.2.32 /30	192.168.2.33	192.168.2.34	192.168.2.35
17	Router Link 8	192.168.2.36 /30	192.168.2.37	192.168.2.38	192.168.2.39
18	Router Link 9	192.168.2.40 /30	192.168.2.41	192.168.2.42	192.168.2.43
19	Router Link 10	192.168.2.44 /30	192.168.2.45	192.168.2.46	192.168.2.47
20	Router Link 11	192.168.2.48 /30	192.168.2.49	192.168.2.50	192.168.2.51

Backups (Eth.Channel & HSRP)

HSRP

- **HSRP** (**H**ot **S**tandby **R**outer **P**rotocol) is : Cisco protocol used to provide router **redundancy** in a network.

- It allows two or more routers to work together so that <u>if the active router fails</u>, another router can **take over** without interrupting network traffic.

- One router is elected as the **active** router, and another as the **standby** router.

- Uses a **virtual IP address** shared between routers, which is configured on hosts as their default gateway.

- Ensures **high availability** and continuous network connectivity for critical devices.

Backups (Eth.Channel & HSRP)

HSRP

- **Configuration Example :**

Active router case (higher priority) :

```
interface GigabitEthernet0/0
 no ip address
 duplex auto
 speed auto
!
interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 192.168.1.2 255.255.255.0
 ip helper-address 192.168.10.10
 ip access-group Block out
 standby 2 ip 192.168.1.1
 standby 2 priority 150
 standby 2 preempt
!
interface GigabitEthernet0/0.20
 encapsulation dot1Q 20
 ip address 192.168.2.2 255.255.255.0
 ip helper-address 192.168.10.10
 ip access-group Block out
 standby 2 ip 192.168.2.1
 standby 2 priority 150
 standby 2 preempt
!
```

Backups (Eth.Channel & HSRP)

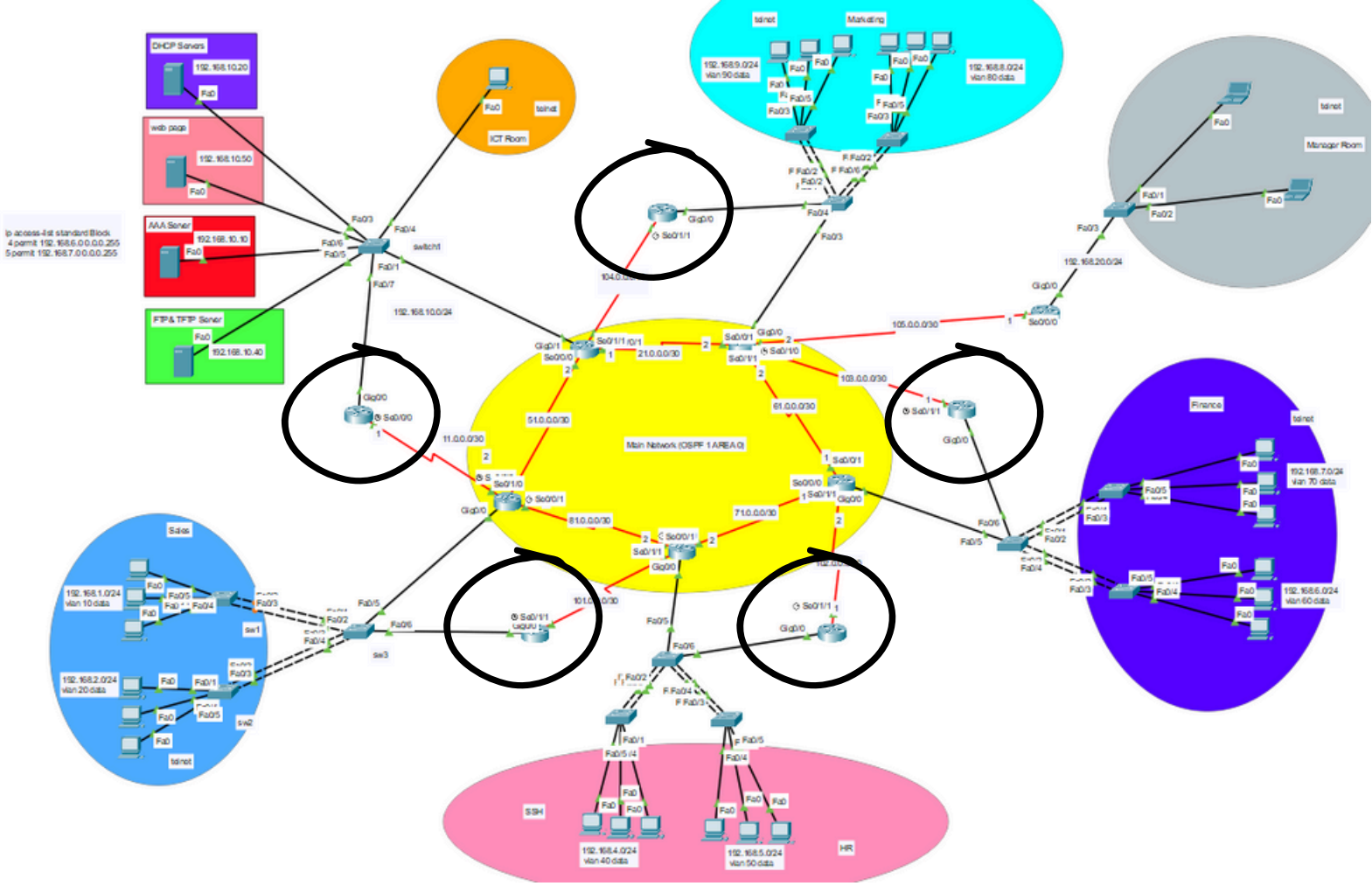
HSRP

- **Configuration Example :**

Standby router case (lower priority) :

```
interface GigabitEthernet0/0
 no ip address
 duplex auto
 speed auto
!
interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 192.168.1.3 255.255.255.0
 ip access-group Block out
 standby 2 ip 192.168.1.1
 standby 2 preempt
!
interface GigabitEthernet0/0.20
 encapsulation dot1Q 20
 ip address 192.168.2.3 255.255.255.0
 ip access-group Block out
 standby 2 ip 192.168.2.1
 standby 2 preempt
!
```

NOTE :
that we didn't set a priority for the **standby** router so it took the **default value** (100)



Ellipses are represent the locations of used **Standby Routers** in our project

Extra Improvements

Port Security

- **Port security** is a Layer 2 feature used on network switches to control access to a port based on the MAC address of devices. It helps protect the network by preventing unauthorized devices from connecting. Administrators can specify which MAC addresses are allowed or set limits on how many can connect. If a violation occurs, the switch can take actions such as shutting down the port or restricting traffic. Port security is commonly used in LAN environments to enhance security on access ports. It's especially useful in preventing MAC flooding and unauthorized access.

- **In our project** , we used it to protect our servers from attacks and prevent other users from acting as our server

- we useed the static method port security that tiea a mac address to a port

Extra Improvements

Port Security

- **Configuration Example :**

```
interface FastEthernet0/1
 switchport mode access
!
interface FastEthernet0/2
 switchport mode access
 switchport port-security
 switchport port-security violation restrict
 switchport port-security mac-address 0090.0CB8.C4B9
!
interface FastEthernet0/3
 switchport mode access
 switchport port-security
 switchport port-security violation restrict
 switchport port-security mac-address 000A.41E7.1D1C
!
interface FastEthernet0/4
 switchport mode access
 switchport port-security
 switchport port-security violation restrict
 switchport port-security mac-address 0001.42CD.32B8
!
interface FastEthernet0/5
 switchport mode access
 switchport port-security
 switchport port-security violation restrict
 switchport port-security mac-address 00D0.BC42.E817
!
interface FastEthernet0/6
 switchport mode access
 switchport port-security
 switchport port-security violation restrict
 switchport port-security mac-address 0004.9ABD.7D2E
!
```

NOTE :

the restrict keyword means prevent the attack and report a log to the administrator

Extra Improvements

Port Address Translation (PAT)

T

- **Port Address Translation (PAT),** also known as NAT overload, is a feature that allows multiple devices on a local network to share a single public IP address. It works by assigning each connection a unique port number, enabling many internal hosts to communicate with external networks simultaneously. PAT modifies both the IP address and port number in outgoing packets. When responses return, the router uses the port information to forward them to the correct internal device. This conserves public IP addresses and adds a layer of security by hiding internal IPs.

- **In our project** , we used PAT to give the manager's network the access on all devices in our system without anyone being able to access devices in manager's network .
- This helped us provide the strong accessibility and security needed to the manager

Extra Improvements

Port Address Translation (PAT)

- **Configuration Example :**

```
access-list 10 permit 192.168.20.0 0.0.0.255
ip nat inside source list 10 interface Serial0/0/0 overload

!
interface GigabitEthernet0/0
 ip address 192.168.20.1 255.255.255.0
 ip helper-address 192.168.10.20
 ip nat inside
 duplex auto
 speed auto
!

interface Serial0/0/0
 ip address 105.0.0.1 255.255.255.252
 ip nat outside
!
```

Extra Improvements

Routing Protocols Redistribution

Redistribution is the process of sharing routes between different routing protocols (like OSPF, EIGRP, RIP, etc.) on a router that runs more than one protocol — called an ASBR (Autonomous System Boundary Router).
It allows networks using different protocols to communicate by importing routes from one routing domain into another.

- **Configuration :**

```
router eigrp 100
 redistribute ospf 1 metric 1 1 1 1 1
 network 105.0.0.0
!
router ospf 1
 log-adjacency-changes
 redistribute eigrp 100 subnets
 network 21.0.0.0 0.0.0.3 area 0
 network 61.0.0.0 0.0.0.3 area 0
 network 192.168.9.0 0.0.0.255 area 0
 network 192.168.8.0 0.0.0.255 area 0
 network 103.0.0.0 0.0.0.3 area 0
!
```

Extra Improvements

Access Control Lists

Access Control Lists (ACLs) are a set of rules used in networking to control and filter traffic. They define which packets are allowed or denied when entering or leaving a network interface. ACLs can filter traffic based on parameters like source or destination IP address, protocol, or port number. They help enhance network security, manage traffic flow, and can be used to restrict access to certain network resources. There are two main types: standard ACLs and extended ACLs

In our project , we used **Standard ACLs** to deny communication between users in different branches and give the full access on all devices to the Manager and ICT members only .

- **Configuration :**

```
ip access-list standard Block
 permit 105.0.0.0 0.0.0.3
 permit 50.50.50.0 0.0.0.255
 permit 192.168.7.0 0.0.0.255
 permit 192.168.6.0 0.0.0.255
 permit 192.168.10.0 0.0.0.255
 permit 192.168.20.0 0.0.0.255
 deny any
```

Extra Improvements & BOUNS

Access Control Lists

Configuration Examble :

```
interface GigabitEthernet0/0.60
 encapsulation dot1Q 60
 ip address 192.168.6.3 255.255.255.0
 ip helper-address 192.168.10.10
 ip helper-address 192.168.10.20
 ip access-group Block out
 standby 4 ip 192.168.6.1
 standby 4 priority 150
 standby 4 preempt
!
interface GigabitEthernet0/0.70
 encapsulation dot1Q 70
 ip address 192.168.7.3 255.255.255.0
 ip helper-address 192.168.10.10
 ip helper-address 192.168.10.20
 ip access-group Block out
 standby 4 ip 192.168.7.1
 standby 4 priority 150
 standby 4 preempt
!
```

NOTE:

- Users in a certain VLAN **can** ping other users in another VLAN in case they are in the same branch through ROAS (ex. Sales)
- Users in a certain branch **can't** access users in another branch .
- Only users in the **ICT** or **Manager's office** has the **full** access.